

Skin Lesion Classification: Melanoma vs Nevus

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Abstract

We study deep learning methods for classifying dermoscopy images as either melanoma (MEL) or melanocytic nevus (NV), using a balanced subset of the ISIC 2018 dataset. We tested five experiments: a baseline CNN, batch normalisation, residual connections, data augmentation, and transfer learning with a frozen VGG-16 network. Transfer learning gave the best validation accuracy (86.3%), closely followed by the residual CNN (85.6%). The VGG-16 transfer learning model was evaluated on the test set, achieving 86.2% accuracy. Residual connections and transfer learning both improved over the 84.0% baseline, but adding data augmentation to the residual model reduced accuracy to 83.5%.

1 Introduction

Skin cancer is one of the most common cancers worldwide. Melanoma is especially dangerous once it spreads, but the National Cancer Institute reports a 99.6% five-year survival rate when caught early. Because skin lesions are visual, machine learning image classification can be used for early detection. We use a balanced subset of the ISIC 2018 dataset (Tschandl et al., 2018), (Codella et al., 2019) with images labelled as either melanoma (MEL, class 0) or melanocytic nevus (NV, class 1). Because the dataset is balanced, accuracy is a fair metric. We tested five models on a fixed validation set of 1,252 images and evaluated our best model on a test set of 1,366 images.

2 Technical Solutions

2.1 Baseline CNN

Our baseline is a simple three-block convolutional network. Each block applies a convolution, a

ReLU activation, and max pooling, followed by two fully-connected layers. No regularisation is used. The images are resized to 128×128 and normalised with ImageNet statistics. This model achieved 84.0% validation accuracy and serves as the reference point for all other experiments.

2.2 Batch Normalisation

We add batch normalisation after each convolutional layer and a Dropout(0.3) layer in the classifier. Batch normalisation stabilises training and allows higher learning rates. This improved validation accuracy to 84.7%, a gain of 0.8 percentage points over the baseline.

2.3 Residual Connections

Residual blocks (He et al., 2016) add a shortcut that skips two convolutional layers: $\text{output} = F(x) + x$. When the number of channels changes, a 1×1 convolution is used to match dimensions. These shortcuts help gradients flow through the network during training, and allow each layer to focus on small corrections rather than learning everything from scratch. This gave validation accuracy of 85.6%, an improvement of 1.7 percentage points over the baseline and the second-best result overall.

2.4 Data Augmentation

During training we apply random horizontal and vertical flips, rotations up to $\pm 20^\circ$, and colour jitter (brightness, contrast, saturation $\pm 20\%$, hue $\pm 5\%$). These are clinically good because lesion orientation and colour balance are not part of the diagnostic. Augmentation is only applied during training and validation and test sets use the standard transform. Surprisingly, adding augmentation to the residual model reduced validation accuracy to 83.5%, below the baseline. We think that this is because the dataset is large enough that the added noise outweighs the regularisation benefit.

2.5 Transfer Learning (VGG-16)

We use VGG-16 (Simonyan and Zisserman, 2014) pre-trained on ImageNet as a frozen feature extractor. All VGG-16 weights are kept fixed throughout training, so no gradient updates are applied to the convolutional layers. The features from the average pooling layer ($512 \times 7 \times 7 = 25,088$ values per image) are extracted once for each split and saved to disk, avoiding repeated inference and significantly reducing computation time. The extracted features are standardised before being passed to the classifier, that is important for stable convergence of the MLP. A two-layer MLP with hidden layers of sizes 512 and 128 is then trained on these features. The VGG-16 input requires its own preprocessing transform (resize to 224×224 , ImageNet normalisation), which is applied separately from the 128×128 transform used by the custom CNNs.

3 Experiments and Results

3.1 Validation Results

Table 1 reports validation accuracy for all five experiments. We can see that Transfer learning (VGG-16) achieves the best validation accuracy of 86.3%, outperforming the baseline by 3.0 pp. Among models trained from scratch, residual connections (+2.55 pp) outperform batch normalisation alone (+1.43 pp). Surprisingly, adding data augmentation to the residual model slightly reduced validation accuracy.

Model	Val Acc
1. Baseline CNN	0.8395
2. CNN + Batch Normalisation	0.8474
3. CNN + Residual Connections	0.8562
4. Residual CNN + Data Aug	0.8347
5. Transfer Learning (VGG-16)	0.8626

Table 1: Validation set accuracy for all experiments.

3.2 Test Set Results

Since the Transfer Learning model is the one that has scored best in the validation set, is the one that we choose to do the evaluation of the test set. That model achieves a test accuracy of 86.16%, closely matching its validation accuracy (86.26%), which indicates low overfitting to the validation set.

Table 2 presents the metrics obtained by the VGG-16 transfer learning model on the test set

(683 MEL, 683 NV), achieving an overall test accuracy of 86.2%. Notably, both classes yield identical precision, recall, and F1-score of 0.86, indicating that the model is equally calibrated for both classes and does not exhibit a bias toward either melanoma or nevus. This symmetry is not optimal because in a medical context, the false-negative rate for melanoma (the more clinically dangerous error) should be lower compared to the false-positive rate.

The confusion matrix in Figure 1 confirms this balance. The model correctly classifies a very similar proportion of both MEL and NV samples, the row-normalised matrix shows that per-class recall is approximately 86% for both categories. Also, note that the number of missclassified images are 94 in the case of False Negative and 95 in the case of the False Positives, confirming that the model is unbiased.

Figure 2 shows a sample of predictions on the test set. The majority of images are classified correctly. The misclassified samples tend to involve marks with ambiguous visual characteristics that could result challenging even for clinical experts.

4 Conclusion

We compared 5 deep learning approaches for melanoma vs nevus classification. Transfer learning with VGG-16 achieved the best validation accuracy (86.3%), demonstrating strong cross-domain transfer from ImageNet to dermoscopy images. Among custom-trained models, residual connections provided the largest single improvement over the baseline. Data augmentation did not help in combination with residual connections in this setting, though it is beneficial in principle and may help with larger data variability. Our best custom model reached 85.6% validation accuracy with balanced precision and recall across both classes. On the held-out test set, the VGG-16 model achieved an accuracy of 86.15%, with perfectly balanced precision and recall of 0.86 for both classes, confirming that the strong validation performance generalises well and that the model does not exhibit a systematic bias toward either class.

4.1 Limitations

Several limitations affect real-world use. First, the task only includes two classes, while real clinical systems would need to classify more lesion types.

Table 2: **Per-class evaluation metrics for the best combined model.** The table reports precision, recall, F1-score, and support for each class on the test set. The dataset is balanced, with 683 samples per class, and the macro average summarises overall performance across classes.

Class / Avg	Precision	Recall	F1-score	Support
MEL	0.86	0.86	0.86	683
NV	0.86	0.86	0.86	683
Macro avg	0.86	0.86	0.86	1366

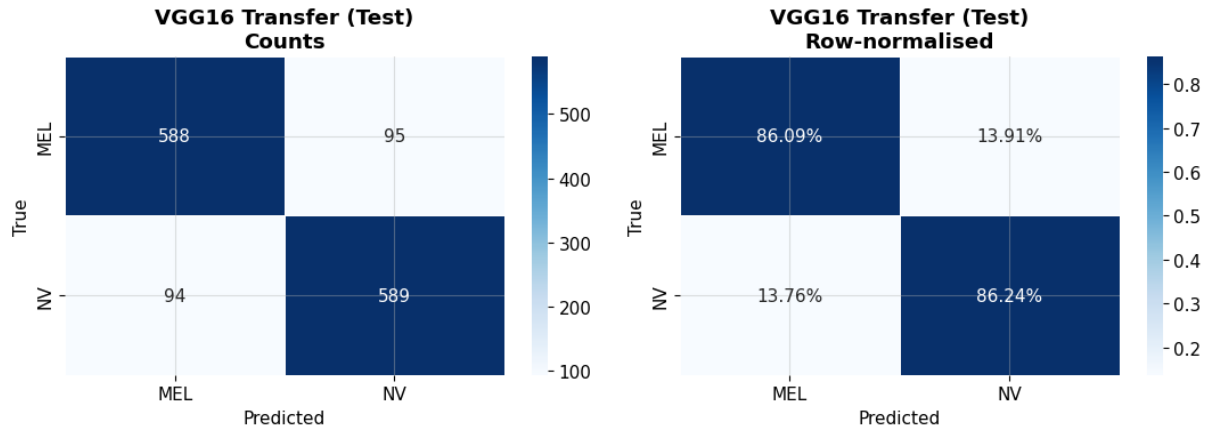


Figure 1: Confusion matrices for the VGG16 transfer learning model on the test set, showing raw counts and normalised results. The model achieves balanced recall for both classes, with 86.09% for MEL and 86.24% for NV.

Second, the custom CNNs use 128×128 images, which may lose important visual details. Third, accuracy is not the best metric for medical diagnosis because missing a melanoma is more serious than a false alarm. Our best model misses 13.91% of melanoma cases, so improving melanoma recall would be more appropriate. Finally, the dataset mainly comes from Austrian and Australian hospitals, so the model may not generalise well to all skin tones or imaging conditions.

4.2 Ethical Considerations

The use of AI for skin lesion classification raises serious ethical concerns. Machine learning models can be very useful, but they should be used as support tools and not as the main basis for diagnosis. Medical experts should supervise model performance, validate the results, and ensure that these systems are accurate and reliable across different skin tones. It is important to address biases in the training data, since models that are not tested on diverse populations may perform worse for underrepresented groups. AI tools in dermatology should be implemented carefully, with human oversight, transparency, and continuous evaluation.

References

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Transfer Learning (VGG-16) — Test Set Predictions

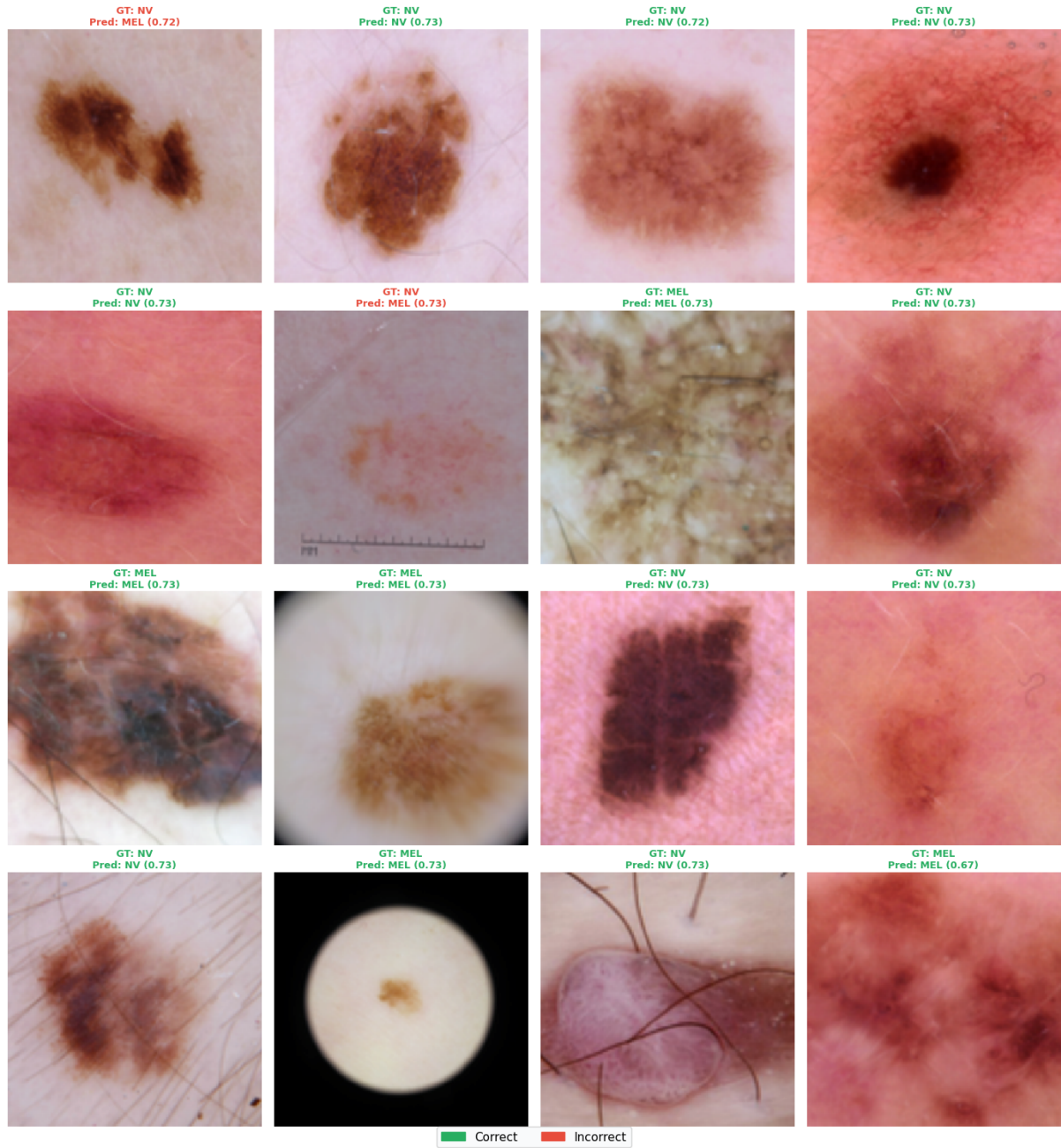


Figure 2: Example test predictions from the VGG16 transfer learning model, showing dermoscopy images with ground-truth and predicted labels. Green labels indicate correct classifications, while red labels indicate incorrect predictions.